# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS**
**COURSE CODE: CSA0713**

---

**COURSE OUTCOME COVERED**

***CO2:*** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

*Assessment Tool Weightage: 30%*

---

Annexure A

# Scenario-Based

---

***Code of Conduct:***

*I C. PRIYANKA / (192511074 )  (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: C. PRIYANKA***

Annexure A

# Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

**2001:0db8:abcd:0000:0000:0000:0000:0000/64**

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

**192.168.2.45**

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANSWERS

**1) 1. Network Address and Broadcast Address**

- School of Computing (/25): Network – **192.168.100.0**, Broadcast – **192.168.100.127**
- School of Electronics (/26): Network – **192.168.100.128**, Broadcast – **192.168.100.191**
- School of Mechanical Engineering (/27): Network – **192.168.100.192**, Broadcast – **192.168.100.223**

**2. Valid Host IP Address Range**

- School of Computing: **192.168.100.1 – 192.168.100.126**
- School of Electronics: **192.168.100.129 – 192.168.100.190**
- School of Mechanical Engineering: **192.168.100.193 – 192.168.100.222**

**3. Usable Host Addresses**

- School of Computing: **126 hosts**
- School of Electronics: **62 hosts**
- School of Mechanical Engineering: **30 hosts**

**4. Remaining Unused IP Address Range**

- **192.168.100.224 – 192.168.100.255 (/27)**
- Usable Hosts: **192.168.100.225 – 192.168.100.254**
- Broadcast Address: **192.168.100.255**

**2) 1. Network Address and Broadcast Address**

- **Software Development (/24):** Network Address – **10.10.0.0**, Broadcast Address – **10.10.0.255**
- **Quality Assurance (/25):** Network Address – **10.10.1.0**, Broadcast Address – **10.10.1.127**
- **Human Resources (/26):** Network Address – **10.10.1.128**, Broadcast Address – **10.10.1.191**
- **Executive Management (/27):** Network Address – **10.10.1.192**, Broadcast Address – **10.10.1.223**

**2. Valid Host IP Address Range**

- Software Development: **10.10.0.1 – 10.10.0.254**
- Quality Assurance: **10.10.1.1 – 10.10.1.126**
- Human Resources: **10.10.1.129 – 10.10.1.190**
- Executive Management: **10.10.1.193 – 10.10.1.222**

**3. Number of Usable Hosts**

- Software Development: **254 hosts**
- Quality Assurance: **126 hosts**
- Human Resources: **62 hosts**
- Executive Management: **30 hosts**

**4. Remaining Unused IP Address Range**

- **Unused IP Range: 10.10.1.224 – 10.10.255.255**

**3) 1. Compressed IPv6 Address**

**2001:db8:abcd::/64**

**2. Expanded IPv6 Address**

**2001:0db8:abcd:0000:0000:0000:0000:0000/64**

**3. Network Prefix**

**2001:db8:abcd::/64**

**4. Total Number of Host Addresses Supported**

**$2^{64}$ = 18,446,744,073,709,551,616 host addresses**

**5. Total Number of Possible Host Addresses**

**18,446,744,073,709,551,616 ($2^{64}$) addresses**

**Conclusion:** The hospital's IPv6 **/64** network provides a unique network prefix and supports **18,446,744,073,709,551,616** possible host addresses, making it suitable for connecting thousands of medical devices and IoT sensors.

**4) 1. Destination Subnet**

**192.168.2.0/24**

**2. Matching Routing Table Entry**

**192.168.2.0/24**

**3. Next-Hop Network/Interface**

**Interface connected to the 192.168.2.0/24 network.**

**4. Default Route**

**0.0.0.0/0 (Default Gateway)**

**Conclusion**

The destination IP address **192.168.2.45** belongs to the **192.168.2.0/24** subnet. The router matches the routing table entry **192.168.2.0/24** and forwards the packet through the interface connected to that network. If no matching route exists, the router uses the **default route (0.0.0.0/0)** to forward the packet.

5) **1. Network Address and Broadcast Address**

- **Administration LAN:** Network Address – **192.168.10.0**, Broadcast Address – **192.168.10.63**
- **Finance LAN:** Network Address – **192.168.10.64**, Broadcast Address – **192.168.10.127**

**2. Valid Host IP Address Range**

- **Administration LAN: 192.168.10.1 – 192.168.10.62**
- **Finance LAN: 192.168.10.65 – 192.168.10.126**

**3. Suitable IP Addresses for Three Computers**

**Administration LAN**

- Computer 1: **192.168.10.2**
- Computer 2: **192.168.10.3**
- Computer 3: **192.168.10.4**

**Finance LAN**

- Computer 1: **192.168.10.66**
- Computer 2: **192.168.10.67**
- Computer 3: **192.168.10.68**

**4. Default Gateway**

- **Administration LAN: 192.168.10.1**
- **Finance LAN: 192.168.10.65**

**Conclusion**

The **Administration LAN (192.168.10.0/26)** and **Finance LAN (192.168.10.64/26)** are correctly subnetted with separate network and broadcast addresses. Hosts in each LAN use the assigned **default gateway** (**192.168.10.1** and **192.168.10.65**) to communicate with other networks through the Layer-3 router.

## RUBRICS FOR CALCULATION

| Scenario Based Assessment | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage (%)** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |